

Math Review and Supplement

1 Math Review

1.1 Points and vectors in Euclidean n-space

Definition 1.1 *1 dimensional Euclidean space: the set of real numbers R*

Definition 1.2 *Euclidean 2-space /Cartesian plane:*

$$R \times R = R^2 = \{(x_1, x_2) | x_1 \in R, x_2 \in R\}$$

Definition 1.3 *Euclidean n-space:*

$$R^n = \{(x_1, x_2, \dots, x_n) | x_i \in R, i = 1, 2, \dots, n\}$$

Definition 1.4 *Elements in R^n are called "vectors"*

Note 1.1 *for $x, y \in R^n$*

- $x = y$ if $x_i = y_i, i = 1, 2, \dots, n$
- $x \geq y$ if $x_i \geq y_i, i = 1, 2, \dots, n$
- $x \gg y$ if $x_i > y_i, i = 1, 2, \dots, n$
- $x > y$ if $x \geq y$ and $x \neq y$
- $x \leq y, x < y, x \ll y$ are similarly defined

Note 1.2 $R_+^n = \{x \in R^n | x \geq 0\}$

Note 1.3 $R_{++}^n = \{x \in R^n | x \gg 0\}$

Definition 1.5 *The algebra of vectors*

*given $x = (x_1, x_2, \dots, x_n) \in R^n, y = (y_1, y_2, \dots, y_n) \in R^n, r \in R$
then*

$$x + y = (x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

$$r \cdot x = (rx_1, rx_2, \dots, rx_n)$$

Definition 1.6 *Length*

Let $x \in R^n$, the length of x (the Euclidean norm of x) denoted $\|x\|$ is denoted as

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

Definition 1.7 *Distance*

Let $x, y \in R^n$, the Euclidean distance between x and y , denoted $d(x, y)$, is defined as norm of $x - y$

$$d(x, y) = \|x - y\| = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}$$

- symmetric in x and y : $d(x, y) = d(y, x)$
- relies on geometric interpretation of $x - y$ as displacement between x and y
- in general, distance measures are called metrics

Definition 1.8 *Inner product*

Let $x, y \in R^n$, the Euclidean inner product (or dot product) of x and y , denoted $x \cdot y$ is defined as

$$x \cdot y = \sum_{i=1}^n x_i \cdot y_i$$

Note 1.4 Let $u, v \in R^n$, let θ be the angle between them, then

- $u \cdot v = \|u\| \|v\| \cos \theta$
- $\|u + v\| \leq \|u\| + \|v\|$
- $|\|u\| - \|v\|| \leq \|u - v\|$

Definition 1.9 *Open set*

A set S in R^m is open if for each $x \in S$, there exists an open ϵ -ball around x completely contained in S

$x \in S \implies$ there is an $\epsilon > 0$ such that $B_\epsilon(x) \subset S$

- Any union of open sets is open
- The finite intersection of open sets is open

Definition 1.10 *Closed sets*

A set S in R^m is closed if and only if its complement, $S^c = R^m - S$, is open

- Any intersection of closed sets is closed
- The finite union of closed sets is closed

Note 1.5 “iff” means “if and only if”

1.2 Calculus

Definition 1.11 *The derivative*

$f : R \rightarrow R$ differentiable at $x_0 \in R$ if

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists (as a finite number). If the limit exists, its value is called the derivative of f at x_0 , and is denoted $f'(x_0)$ or $\frac{df}{dx}(x_0)$

- Differentiable at $x_0 \implies$ continue at x_0
- $\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$ is finite $\implies \lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] = 0$
- Let $\{h_n\}_{n=1}^\infty$ be a sequence of real number. Differentiability of f at x_0 requires segment of the form $\left\{ \frac{f(x_0 + h_n) - f(x_0)}{h_n} \right\}_{n=1}^\infty$ converge and have the same limit for all $\{h_n\}_{n=1}^\infty \rightarrow 0$
- Discontinuous at $x_0 \implies$ not differentiable at x_0
- Differentiable on a set: differentiable at every point in the set

Theorem 1.1 *Product Rule*

Suppose $f : R \rightarrow R$ and $g : R \rightarrow R$ differentiable at $x_0 \in R$, then the function $(f \cdot g)x = f(x) \cdot g(x)$ is differentiable at x_0 and $(f \cdot g)'x = f'(x) \cdot g(x) + f(x) \cdot g'(x)$

Theorem 1.2 *Fermat's Lemma*

Let $f : R \rightarrow R$ be differentiable, if f attains a "local maximum" or "local minimum" at x_0 , then $f'(x_0) = 0$

Theorem 1.3 *Rolle's Theorem*

Suppose $g : [a, b] \rightarrow R$ is differentiable on $[a, b]$, suppose $g(a) = g(b)$, then there is a point $x_1 \in (a, b)$ such that $g'(x_1) = 0$

Theorem 1.4 *Taylor's Theorem*

Given $f : [a, b] \rightarrow R$ differentiable and $x^* \in [a, b]$, then for each non-negative integer n and for any $h \neq 0$ such that $x^* + h \in [a, b]$, then there exists an x_1 in the open interval bounded by x^* and $x^* + h$ such that

$$f(x^* + h) = \sum_{k=0}^n \frac{f^{(k)}(x^*)}{k!} h^k + \frac{f^{(n+1)}(x_1)}{(n+1)!} h^{n+1}$$

Theorem 1.5 *Chain Rule*

Suppose $f : R \rightarrow R$ and $g : R \rightarrow R$ differentiable, then the function $(f \circ g)(x) = f(g(x))$ is differentiable for each $x_0 \in R$ and $(f \circ g)'(x_0) = f'(g(x_0))g'(x_0)$

Theorem 1.6 *Inverse Function Theorem*

Given $f : [a, b] \rightarrow \mathbb{R}$ differentiable and $f' \neq 0$ for all $x \in [a, b]$, then inverse function $g : f([a, b]) \rightarrow [a, b]$ is differentiable at all $y \in f([a, b])$ with derivative

$$g'(y) = \frac{1}{f'(g(y))}$$

Theorem 1.7 *Fundamental Theorem of Calculus*

If $f : [a, b] \rightarrow \mathbb{R}$ is integrable on $[a, b]$, then $\int_a^b f(x)dx = F(b) - F(a)$ for any antiderivative of $f(\cdot)$, $F(\cdot)$

Theorem 1.8 *Integration by parts*

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ differentiable, then the function $(f \cdot g)x = f(x) \cdot g(x)$ is differentiable and we have

$$\int f'(x) \cdot g(x)dx = f(x) \cdot g(x) + \int f(x) \cdot g'(x)dx$$

Definition 1.12 *Partial differentiation*

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ for $i = 1, 2, \dots, n$. The partial derivative of f with respect to x_i at the point $x_0 = (x_1^0, x_2^0, \dots, x_n^0)$ is denoted by $\frac{\partial f}{\partial x_i}(x_0)$ and defined by

$$\frac{\partial f}{\partial x_i}(x_0) = \lim_{h \rightarrow 0} \frac{f(x_1^0, x_2^0, \dots, x_i^0 + h, \dots, x_n^0) - f(x_1^0, x_2^0, \dots, x_i^0, \dots, x_n^0)}{h}$$

if the limit exists

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable means all n partial derivative exists at the point x .
- $\frac{\partial f}{\partial x_i} : \mathbb{R}^n \rightarrow \mathbb{R}$

Definition 1.13 *Gradient vector*

Suppose $f(x)$ is differentiable at x_0 , then the gradient vector of $f(x)$ at x_0 denoted by $\text{grad}f(x_0)$ is

$$\text{grad} f(x_0) = \left(\frac{\partial f(x_0)}{\partial x_1}, \frac{\partial f(x_0)}{\partial x_2}, \dots, \frac{\partial f(x_0)}{\partial x_n} \right)$$

Theorem 1.9 *Implicit function theorem*

Let $F : \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$ differentiable, $y^* \in \mathbb{R}$ and $x^* \in \mathbb{R}^m$, such that $F(y^*, x^*) = 0$ and $\frac{\partial F}{\partial y}(y^*, x^*) \neq 0$. Then there exist $\varepsilon > 0$ and a function $h : \mathbb{R}^m \rightarrow \mathbb{R}$ differentiable such that $y = h(x)$ and $F(h(x), x) = 0$ for all $x \in \mathcal{B}_\varepsilon(x^*)$. Also for $i = 1, 2, \dots, n$

$$\frac{\partial h}{\partial x_i}(x^*) = -\frac{\frac{\partial F}{\partial x_i}(y^*, x^*)}{\frac{\partial F}{\partial y}(y^*, x^*)}$$

1.3 Linear algebra

Definition 1.14 *Basic matrix operations*

Consider two matrix, $A = [a_{ij}]$, $B = [b_{ij}]$, $i = 1, 2, \dots, k$, $j = 1, 2, \dots, n$

- *Summation*

$$A + B = [a_{ij} + b_{ij}], i = 1, 2, \dots, k, j = 1, 2, \dots, n$$

- *Scalar multiplication*

$$r \cdot A = [r \cdot a_{ij}], i = 1, 2, \dots, k, j = 1, 2, \dots, n$$

- *Transposition*

$$A' = [a_{ji}], i = 1, 2, \dots, k, j = 1, 2, \dots, n$$

- *Matrix multiplication: $A : k \times n$; $B : n \times m$*

$$A \cdot B = \left[\sum_{l=1}^n a_{il} b_{lj} \right], i = 1, 2, \dots, k, j = 1, 2, \dots, n$$

- A^{-1} is the inverse of A such that $A \cdot A^{-1} = I$
- $B/A = B \cdot A^{-1}$

Definition 1.15 *Some special matrix*

- *Square matrix: $A : n \times n$*
- *Diagonal matrix : $a_{ij} = 0$ for all $i \neq j$*
- *Identity matrix : $a_{ij} = 0$ for all $i \neq j$ and $a_{ii} = 1$ for $i = 1, 2, \dots, n$*
- *Symmetric matrix : $a_{ij} = a_{ji}$ for all $i = 1, 2, \dots, n$*

Definition 1.16 *Minor*

Let A_{ij} be the $(n-1) \times (n-1)$ matrix obtained from A by deleting i th row and j th column. Define $M_{ij} = \det A_{ij}$, ij minor of A and is usually written as C_{ij} .

Definition 1.17 *The determinant*

Let A be a $n \times n$ matrix. The determinant of A , $\det A$, is defined by

1. For $n = 1$, $\det A = a_{11}$

2. For $n > 1$, $\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$ for any $i = 1, 2, \dots, n$

where $(-1)^{i+j} M_{ij} = (-1)^{i+j} \det A_{ij}$ is called the (i, j) th cofactor of A

Note 1.6 A square matrix A is nonsingular iff $\det A$ is nonzero

Note 1.7 A square matrix A is invertible iff it is nonsingular

Note 1.8 Let A be a $n \times n$ matrix. Let B be the $n \times n$ matrix formed from A by interchanging any two rows or any two columns, then $\det B = -\det A$

Note 1.9 Let A be a $n \times n$ matrix with two identical rows or two identical columns, then $\det A = 0$.

Definition 1.18 Adjoint matrix

Define adjoint matrix = transpose of cofactor matrix, denoted by A^*

$$A^* = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

- Let A be a $n \times n$ matrix with $\det A \neq 0$ then $A \cdot A^* = \det A \cdot I$

Theorem 1.10 Crammer's rule

Let A be a $n \times n$ matrix with $\det A \neq 0$. The unique solution $x = (x_1, x_2, \dots, x_n)'$ to the system of n linear equation $Ax = b$ is given by

$$x_j = \frac{\det B_j}{\det A} \text{ for } j = 1, 2, \dots, n$$

where B_j is the matrix A by replacing j th column of A with b

Definition 1.19 Quadratic form

A quadratic form is a real valued function on $R^n, Q : R^n \rightarrow R$, of form

$$Q(x) = x'Ax$$

where A is a square symmetric matrix

Definition 1.20 Sign definiteness

Let A be a $n \times n$ symmetric matrix. Then A is

- positive definite if $x'Ax > 0$ for all $x \neq 0, x \in R^n$.
- positive semi-definite if $x'Ax \geq 0$ for all $x \in R^n$.
- negative definite if $x'Ax < 0$ for all $x \neq 0, x \in R^n$.
- negative semi-definite if $x'Ax \leq 0$ for all $x \in R^n$.
- indefinite if $x'Ax > 0$ for some $x \in R^n$, and $x'Ax < 0$ for some $x \in R^n$.

1.4 Probability theory

Definition 1.21 Probability density functions

For any random variable, its probability density function (PDF) shows the probability that each specific outcome will occur.

If we denote all the outcomes for a discrete random variable by x_i for $i = 1, \dots, n$, then we must have:

$$\sum_{i=1}^n f(x_i) = 1$$

For a continuous random variable we have:

$$\int_{-\infty}^{+\infty} f(x) = 1$$

Note 1.10 A few important PDFs

- Binomial distribution

$$\begin{aligned} f(x=1) &= p \\ f(x=0) &= 1-p \end{aligned}$$

where $0 < p < 1$

- Uniform distribution

$$\begin{aligned} f(x) &= \frac{1}{b-a}, \text{ for } a \leq x \leq b \\ f(x) &= 0, \text{ for } x < a \text{ or } x > b \end{aligned}$$

- Exponential distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

- Normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Definition 1.22 Expected value

The expected value of a random variable is the numerical value that the random variable might be expected to have, on average, denoted by $E(x)$

For a discrete random variable that takes on the values $x_1, x_2, \dots, x_n$, the expected value is defined as

$$E(x) = \sum_{i=1}^n x_i f(x_i)$$

For a continuous random variable

$$E(x) = \int_{-\infty}^{+\infty} x f(x) dx$$

Definition 1.23 *Cumulative distribution function*

Cumulative distribution function(CDF) $F(x)$, defined as

$$F(x) = \int_{-\infty}^x f(t)dt$$

$F(x)$ represents the probability that the random variable t is less than or equal to x

Definition 1.24 *Variance*

The variance is defined as the “expected squared deviation” of a random variable from its expected value

$$Var(x) = \sigma_x^2 = E[(x - E(x))^2] = \int_{-\infty}^{+\infty} (x - E(x))^2 f(x)dx$$

2 Concave and quasi-concave functions, convex sets

Definition 2.1 *Convex sets*

A set $U \in R^n$ is convex if for all $u, v \in U$ and for all $h \in [0, 1]$

$$hu + (1 - h)v \in U$$

A set $U \in R^n$ is strictly convex if for all $u, v \in U, u \neq v$, and for all $h \in (0, 1)$, there exists $\varepsilon > 0$ such that

$$B_\varepsilon(hu + (1 - h)v) \subset U$$

- *If U is open and convex, then U is strictly convex.*

Definition 2.2 *Concave/Convex functions*

Let U be a convex subset of R^n , $F : U \rightarrow R$ is concave (convex) if

$$F(hu + (1 - h)v) \geq (\leq) hF(u) + (1 - h)F(v)$$

, for all $u, v \in U$ and for all $h \in [0, 1]$.

Let U be a convex subset of R^n , $F : U \rightarrow R$ is strictly concave (strictly convex) if

$$F(hu + (1 - h)v) > (<) hF(u) + (1 - h)F(v)$$

, for all $u, v \in U, u \neq v$ and for all $h \in [0, 1]$.

Theorem 2.1 *Let U be a convex subset of R^n and $F : U \rightarrow R$. For any $x, y \in U$, define $g_{x,y} : [0, 1] \rightarrow R$ as follows:*

For $t \in [0, 1]$, $g_{x,y}(t) = F(tx + (1 - t)y)$. $g_{(x,y)}(\cdot)$ is called the restriction of $F(\cdot)$ to the line segment between x and y .

Then $F : U \rightarrow R$ is concave if and only if $g_{x,y} : [0, 1] \rightarrow R$ is concave for all $x, y \in U$.

Lemma 2.1 Let I be an interval of R (convex set in R) and let $f : I \rightarrow R$ be differentiable. Then f is concave iff

$$f(y) - f(x) \leq f'(x)(y - x)$$

for all $x, y \in I$

Lemma 2.2 Let I be an interval of R (convex set in R) and $f : I \rightarrow R$ be differentiable and concave. Then $f''(x) \leq 0$ for all $x \in I$

Proposition 2.1 Let U be convex, open and $\subset R^n$. If $F : U \rightarrow R$ is differentiable, then the following are equivalent

- (a) for all $u, v \in U$, and for all $h \in [0, 1]$, $F(hu + (1-h)v) \geq hF(u) + (1-h)F(v)$
- (b) for all $u, v \in U$, $F(u) \leq F(v) + \frac{\partial F}{\partial x}(v)(u - v)$
- (c) $\frac{\partial^2 F}{\partial x^2}(v)$ is negative semi-definite for all $v \in U$

Proposition 2.2 U is convex and $\subset R^n$, $F : U \rightarrow R$ is concave, then $\{x \in U | F(x) \geq a\}$ is convex for all $a \in R$

- $\{x \in U | F(x) \geq a\}$ is called upper contour set.
- If $F : U \rightarrow R$ is concave, then the upper contour set is convex for all $a \in R$

Proposition 2.3 Properties of concave functions

- Critical points, i.e., x^* s.t. $f'(x^*) = 0$ are automatically global maxima
- Weighted sum of concave functions is a concave function
- The level sets of a concave function have the right shapes for consumption and production theory, i.e., ordinal property of concave function.
 - The set above the level set of a concave function is convex
 - Diminishing marginal rate of substitution

Definition 2.3 Quasi-concave functions

Let U be a convex set of R^n , $F : U \rightarrow R$ is quasi-concave if $\{x \in U | F(x) \geq a\}$ is convex for all $a \in R$

Theorem 2.2 Let U be a convex set of R^n , $F : U \rightarrow R$ is quasi-concave if and only if for all $u, v \in U$, and all $h \in [0, 1]$, $F(hu + (1-h)v) \geq \min \{F(u), F(v)\}$

Definition 2.4 Let U be a convex set of R^n , $F : U \rightarrow R$ is strictly quasi-concave if and only if for all $u, v \in U$, $u \neq v$, and all $h \in (0, 1)$, $F(hu + (1-h)v) > \min \{F(u), F(v)\}$

Theorem 2.3 Let U be an open convex set of R^n and $F : U \rightarrow R$ be differentiable. Then the following statements are equivalent

(a) for all $u, v \in U$ such that $F(u) \geq F(v)$ and for all $h \in [0, 1]$,

$$F(hu + (1 - h)v) \geq F(v)$$

(b) for all $u, v \in U$ such that $F(u) \geq F(v)$

$$\frac{\partial F}{\partial x}(v)(u - v) \geq 0$$

(c) F is quasi-concave if and only if $D^2F(x)$ is constrained negative semi-definite, i.e., $zD^2F(x)z \leq 0, \forall z$ such that $DF(x)z = 0$

(d) Define the Bordered matrix

$$B = \begin{pmatrix} 0 & F_1 & F_2 & \cdots & F_k \\ F_1 & F_{11} & F_{12} & \cdots & F_{1k} \\ F_2 & F_{21} & F_{22} & \cdots & F_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ F_k & F_{k1} & F_{k2} & \cdots & F_{kk} \end{pmatrix}$$

Denote B_k k th order successive principal minor of B . If $B_1 < 0, B_2 > 0, \dots, (-1)^n B_n > 0$, then F is quasiconcave.

3 Unconstrained optimization

Definition 3.1 maximization problem

$$\begin{aligned} & \max_x F(x) \\ & \text{subject to } x \in X \end{aligned}$$

Theorem 3.1 Minimize $F(x)$ by $\max -F(x)$
if x^* is a solution to

$$\begin{aligned} & \min_x F(x) \\ & \text{subject to } x \in X \end{aligned}$$

then it is also a solution to

$$\begin{aligned} & \max_x -F(x) \\ & \text{subject to } x \in X \end{aligned}$$

Note 3.1 Elements of a maximization problem

- Objective function: $F : R^n \rightarrow R$ is differentiable.
- Vector of choice variable : $x = (x_1, x_2, \dots, x_n)'$

- Constraint set (or choice set): $X \subset R^n$

Note 3.2 Different forms of X give variety of special cases:

- $X = R^n$, unconstrained optimization
- Equality constrained optimization :

$$X = \{x \in R^n : g_1(x) = b_1, g_2(x) = b_2, \dots, g_m(x) = b_m\}$$

where $g_j : R^n \rightarrow R$ is differentiable and $b_j \in R$ for all $j = 1, 2, \dots, m$,
Expect $n > m$.

- $g_j \leq b_j$ or $x_i \geq 0$.
- Kuhn-Tucker formulation

$$X = \{x \in R^n : x_i \geq 0 \text{ for } i = 1, 2, \dots, n; g_j \leq b_j \text{ for } j = 1, 2, \dots, m\}$$

Definition 3.2 Types of solutions

- x^* is a local solution to maximization problem if there exists $\epsilon > 0$ such that $F(x^*) \geq F(x)$ for all $x \in X \cap B_\epsilon(x^*)$.
- x^* is a strictly local solution to maximization problem if there exists $\epsilon > 0$ such that $F(x^*) > F(x)$ for all $x \in X \cap B_\epsilon(x^*)$ and $x \neq x^*$.
- x^* is a global solution to maximization problem if $F(x^*) \geq F(x)$ for all $x \in X$.
- x^* is a strictly global solution to maximization problem if $F(x^*) > F(x)$ for all $x \in X$ and $x \neq x^*$.

Theorem 3.2 First order necessary condition

if x^* is a local solution to maximization problem with $x \in R^n$, then x^* need to satisfy

$$\frac{\partial F}{\partial x}(x^*) = 0$$

Theorem 3.3 Second order necessary condition

if x^* is a local solution to maximization problem with $x \in R^n$, then x^* need to satisfy $\frac{\partial^2 F}{\partial x^2}(x^*)$ is negative semi-definite.

Note 3.3 Since strict local solution is a subset of local solution, it has the same first-order and second-order necessary conditions.

Theorem 3.4 Sufficient condition

Suppose that we have $x^* \in R^n$ such that $\frac{\partial F}{\partial x}(x^*) = 0$ and $\frac{\partial^2 F}{\partial x^2}(x^*)$ is negative definite, then x^* is a strict local solution to maximization problem with $x \in R^n$.

Theorem 3.5 *Global solution*

- F is concave and $\frac{\partial F}{\partial x}(x^*) = 0$
 $\implies x^*$ is the global solution to maximization problem with $x \in R^n$.
- F is strictly concave and $\frac{\partial F}{\partial x}(x^*) = 0$
 $\implies x^*$ is the strictly global solution to maximization problem with $x \in R^n$.